

Module TH-3: ML workflow and life cycle

Naeemullah Khan

naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

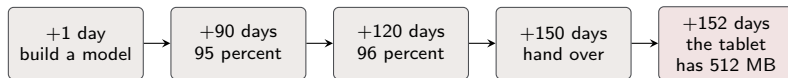
KAUST Academy
King Abdullah University of Science and Technology

October 7, 2026

After this module you can:

- ▶ Name the six stages of an ML project, from the problem definition to the monitoring, and the output of each stage.
- ▶ Explain why a deployment constraint that nobody writes down comes back later as a redesign.
- ▶ Use the feedback of the monitoring stage to decide which earlier stage to change.

A project that failed at the last step

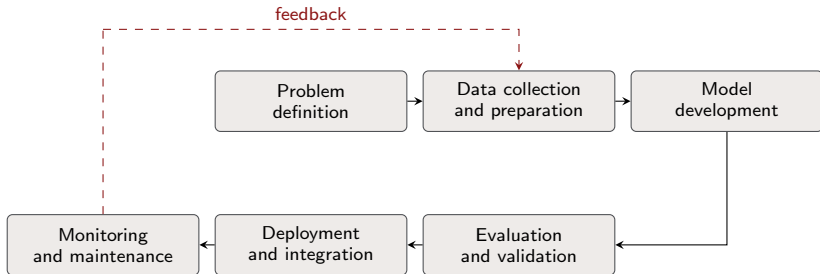


- ▶ The model needs 4 GB of memory. The tablets of the mobile clinics have 512 MB. Five months of work is discarded.
- ▶ The accuracy was good. The skills of the team were good. The failure was a workflow failure.

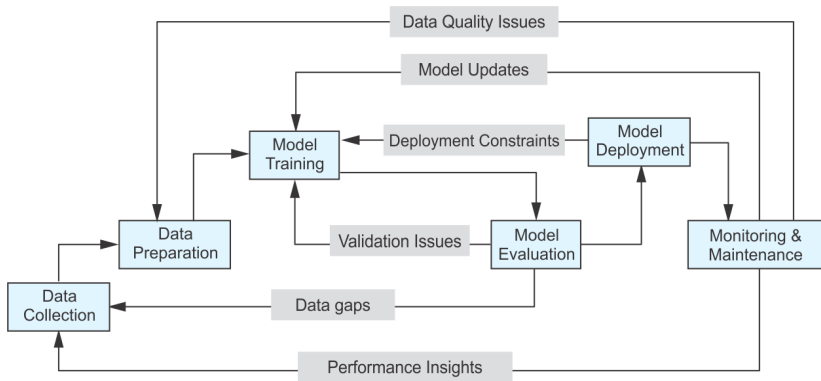
- ▶ A structure for the work, not a method for one model. It names the stages of a project and the output of each stage.
- ▶ It is iterative. A stage sends its result back to an earlier stage as often as it needs to.
- ▶ It covers the running system. Monitoring is a stage of the project, not the end of it.

Key point: The lifecycle is a loop, not a list.

The six stages



Every feedback path has a name



The grey boxes name what travels back. Data gaps send the team to the collection. Deployment constraints go back to the training. Performance insights from the running system decide what happens next.

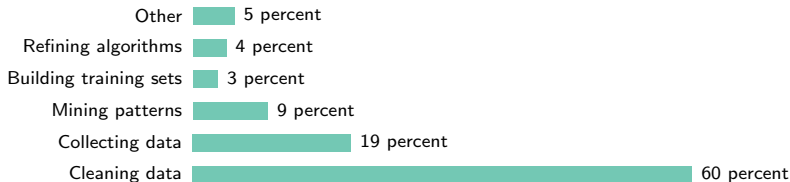
Source: Figure adopted from Volume I, chapter 3.

The output of each stage

Stage	Output that the next stage needs
Problem definition	Targets, the deployment device, the budgets
Data collection	A versioned dataset, the preparation code
Model development	Weights, the training configuration
Evaluation	Metrics per subgroup, the list of failure modes
Deployment	The running system, the instruments, the rollback
Monitoring	The alerts, the review, the incident reports

Each stage is complete when its output exists and passes its check. A stage that ends without a check gives the next stage a defect.

Where the engineering time goes



- ▶ 79 percent of the reported time goes to collecting and cleaning data. 16 percent goes to model work.
- ▶ Data engineering is not preparation. It is most of the project.

Source: CrowdFlower, *Data Science Report 2016*, as reported in Volume I, chapter 3.

ML is not software with a model

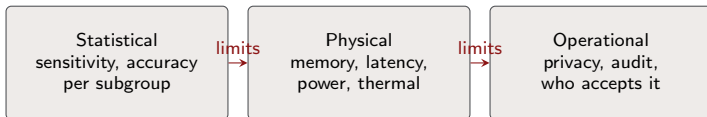
	Written software	A learned system
Specification	Exact functions	A statistical behaviour
Test	One expected result	A distribution with uncertainty
Deployment	Static until an update	Changes when the data changes
Maintenance	Fix the code	Watch the data, then retrain

The behaviour of a learned system comes from the training data. Changing the data changes the behaviour with no change in the code.

- ▶ The behaviour to reach, as a number that a test can check. An average accuracy hides a subgroup that fails.
- ▶ The device and its budgets: memory, latency, power, and the network.
- ▶ The operating rules: privacy, audit, and who accepts the result.

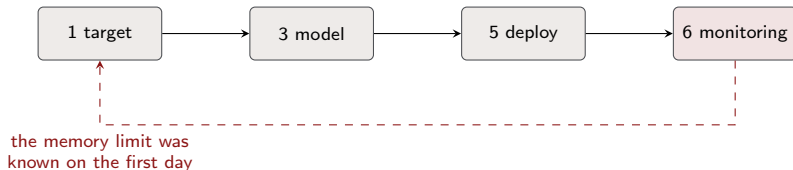
Key point: Write the constraint before the architecture.

The three layers of constraints



A constraint in one layer removes options in the other two. A model that is too large for the device pushes the work back to the architecture, and a missing audit rule can force a retraining later.

Constraints propagate backward



A limit that nobody writes in the problem definition does not disappear. It comes back at the stage that meets it, and it invalidates every decision that was made after the day it should have been known.

Source: Principle of Volume I, chapter 3.

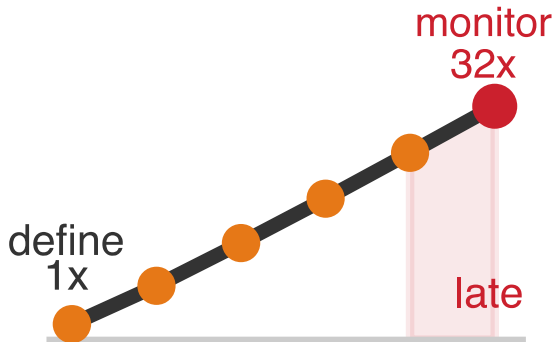
$$\text{correction cost} \approx 2^{N-1} \times \text{stage 1 cost}$$

Found at stage	2	3	4	5	6
Multiple of the stage 1 cost	2	4	8	16	32

This model is an illustration, not a measured law. It shows the shape of the effect: each earlier stage that the correction passes through adds its own validation and testing work.

Source: Illustrative model of Volume I, chapter 3, after Boehm (1981).

The cost curve of a late constraint



The curve is not straight. Each stage that the correction passes through adds its own validation and testing, so the curve steepens. A limit found at monitoring costs 32 times the same limit found at the problem definition.

Source: Figure adopted from Volume I, chapter 3.

- ▶ The targets of stage 1 become data requirements. A model that must run on the kit needs inputs that the kit produces.
- ▶ The dataset has a version. Without it no earlier result can be reproduced.
- ▶ Preparation is code, so it changes, so it needs a test.

A screening system of this kind needs images on the order of 10^5 , each read by several specialists, because one reader disagrees with another on the border cases.

Source: Order of magnitude of Volume I, chapter 3.

- ▶ A screening model reached the accuracy of a specialist in the laboratory.
- ▶ In the clinics of Thailand, 393 of 1 838 submitted images (21 percent) failed the image quality check of the system.
- ▶ The cause was the room, the camera, and the practice of the clinic. No change in the model would have helped.

Key point: A dataset that matches the laboratory does not match the site.

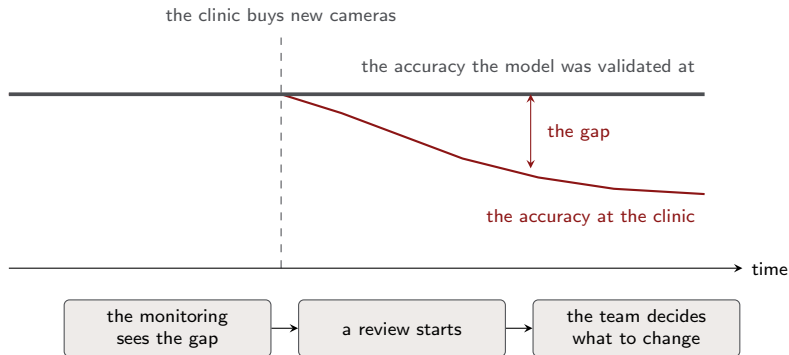
Source: Beede and others (2020), as reported in Volume I, chapter 3.

- ▶ The architecture is chosen inside the budget of stage 1, not after it.
- ▶ MobileNetV2 has 3 504 872 parameters. In `float32` the weights alone need 13.4 MB. A tablet with 512 MB can hold the model, but the input pipeline and the runtime still need room.
- ▶ The training configuration is an output of this stage. Without it the weights cannot be rebuilt.

- ▶ One number is not enough. Report the metric per subgroup that the problem definition named.
- ▶ Fix the threshold here. The threshold is part of the system, so it needs a test.
- ▶ Test on data that no stage has touched, and keep it that way.
- ▶ Test on the device. A model that passes on a laptop can fail on the target with a different number format.

- ▶ The model is now one part of a running system. The path from the sensor to the result decides the latency.
- ▶ Write down what the system promises: latency, uptime, and the behaviour when a stage fails.
- ▶ Keep a way back. A version that fails in the field must be replaceable by the version before it.
- ▶ Check the whole path on the device. A model that runs alone is not a system.

Stage 6: monitoring and maintenance

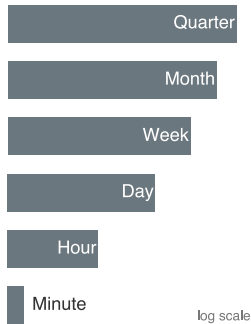


No line of code changed, so nothing failed and no error was raised. The gap between the two lines is the whole reason that stage 6 is a stage of the project.

Review	Period	Finds
Live monitoring	Seconds	A stopped service, a sensor that fails
Update review	Days	Drift, a fall in a metric, a new subgroup
Architecture review	Months	A budget that no longer holds

One loop for all three does not work. A fast loop that waits for the slow reviews reacts too late. A slow loop that runs on every alert spends its budget with no signal.

The span of the review periods



- ▶ The periods of a real system span five orders of magnitude, from a minute to a quarter.
- ▶ The diagram uses a logarithmic scale. Each bar is one period.
- ▶ A drift signal needs a day loop. A change of the budget needs a quarter loop.

Source: Figure adopted from Volume I, chapter 3.

Every stage has an input, an output, and a condition that the output must satisfy. The team checks the condition at the door of the next stage, while the fix is still one change.

Contract	Example of a condition
Problem definition	Every target has a number
Data collection	The data matches the site
Model development	The model fits the budget
Evaluation	No named subgroup is below its floor
Deployment	The latency and the uptime hold
Monitoring	An alert starts a review

- ▶ **An ML project can follow a software plan.** The targets move when the data shows a gap, so the plan moves with them.
- ▶ **Data preparation is a step that ends.** The data of the site changes after the launch, so preparation is a running pipeline.
- ▶ **A good model means a good system.** The number of the model says nothing about the latency, the recovery, or the cost of the device.

Key point: Optimising one stage moves the failure. It does not remove it.

1. Six stages, and an output that the next stage checks.
2. Three layers of constraints: statistical, physical, operational.
3. A constraint that nobody writes down returns later, and it returns as a redesign.
4. Monitoring is a stage of the project. It sends work back to the data and to the model.
5. Every feedback loop has its own speed and its own review.

A team finds during monitoring (stage 6) that its model fails for the images of one clinic. The limit of that clinic was never written in the problem definition.

1. Give the cost multiplier of the illustrative model.
2. Name the stages that the team must revisit.

Work in pairs for 5 minutes.

1. The constraint is found at stage 6, so the multiplier is $2^{6-1} = 32$ times the cost of the same correction at stage 1.
2. Stages 2 to 5: more data for that clinic, a new model, a new evaluation, and a new version on the devices.

The 32 is an illustration, not a measurement. The part that carries over to real work is the list of stages: the correction of a late constraint passes through every stage between.

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.